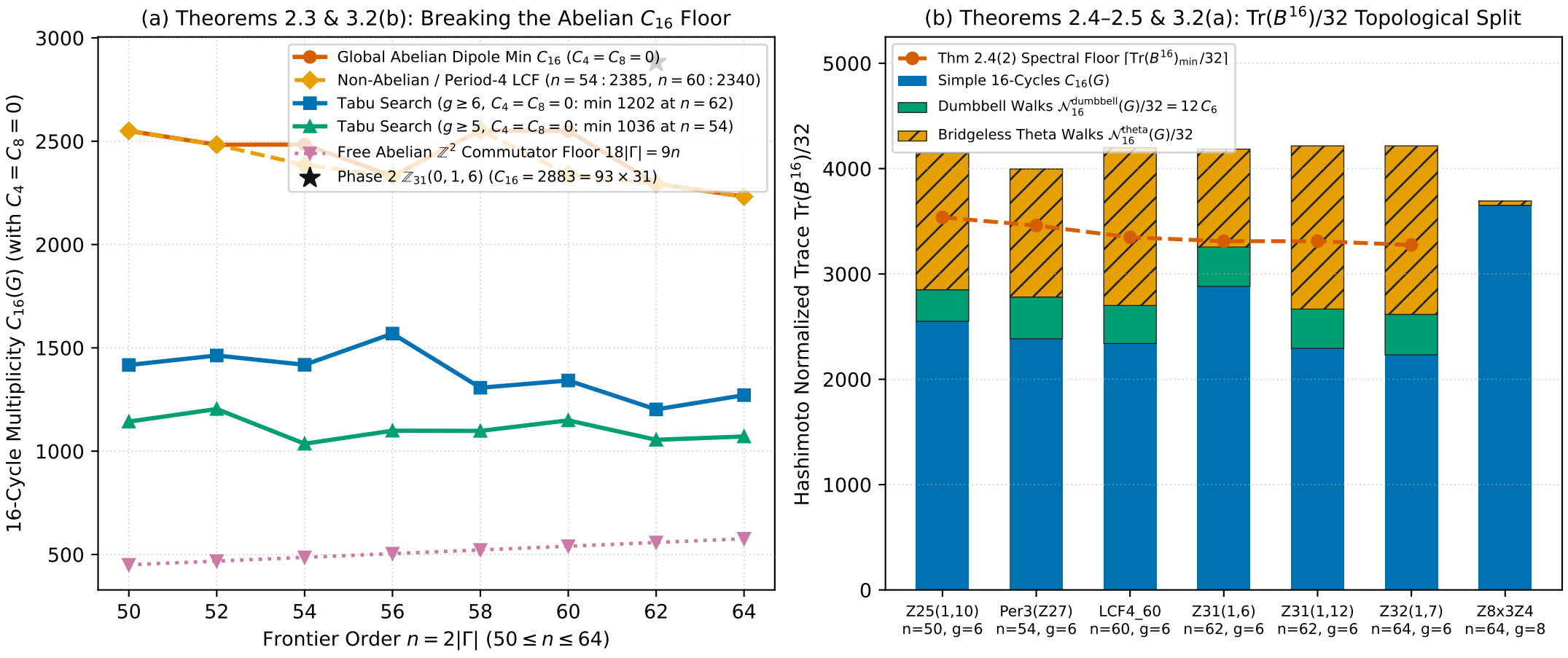
(a) Theorems 2.3 & 3.2(b): Breaking the Abelian C_{16} Floor
16-Cycle Multiplicity $C_{16}(G)$ (with $C_4 = C_8 = 0$)
Frontier Order $n = 2|\Gamma|$ $(50 \le n \le 64)$
Global Abelian Dipole Min C_{16} ($C_4 = C_8 = 0$)
Non-Abelian / Period-4 LCF ($n = 54 : 2385$, $n = 60 : 2340$)
Tabu Search ($g \ge 6$, $C_4 = C_8 = 0$: min 1202 at $n = 62$)
Tabu Search ($g \ge 5$, $C_4 = C_8 = 0$: min 1036 at $n = 54$)
Free Abelian $\mathbb{Z}^2$ Commutator Floor $18|\Gamma| = 9n$
Phase 2 $\mathbb{Z}_{31}(0, 1, 6)$ ($C_{16} = 2883 = 93 \times 31$)
(b) Theorems 2.4–2.5 & 3.2(a): Tr(B^{16})/32 Topological Split
Hashimoto Normalized Trace Tr(B^{16})/32
Thm 2.4(2) Spectral Floor $\lceil \mathrm{Tr}(B^{16})_{\min}/32 \rceil$
Simple 16-Cycles $C_{16}(G)$
Dumbbell Walks $\mathcal{N}_{16}^{\mathrm{dumbbell}}(G)/32 = 12\,C_6$
Bridgeless Theta Walks $\mathcal{N}_{16}^{\mathrm{theta}}(G)/32$
Z25(1,10) n=50, g=6
Per3(Z27) n=54, g=6
LCF4_60 n=60, g=6
Z31(1,6) n=62, g=6
Z31(1,12) n=62, g=6
Z32(1,7) n=64, g=6
Z8x3Z4 n=64, g=8